

Universidad Central del Ecuador
Facultad de Filosofía Letras y Ciencias de
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Curso: 5to "B"

Fecha: 13/05/2026

Asignatura: Física Aplicada

• Grupo 1

$$\vec{A} = 3\vec{i} + 6\vec{j} + 4\vec{k}$$

$$\vec{B} = 5\vec{i} + 2\vec{j} + \vec{k}$$

Módulo A

$$|\vec{A}| = \sqrt{3^2 + 6^2 + 4^2}$$

$$|\vec{A}| = \sqrt{9 + 36 + 16}$$

$$|\vec{A}| = \sqrt{61} = 7.81 \text{ [m]}$$

Módulo B

$$|\vec{B}| = \sqrt{5^2 + 2^2 + 1^2}$$

$$|\vec{B}| = \sqrt{25 + 4 + 1}$$

$$|\vec{B}| = \sqrt{30} = 5.47 \text{ [m]}$$

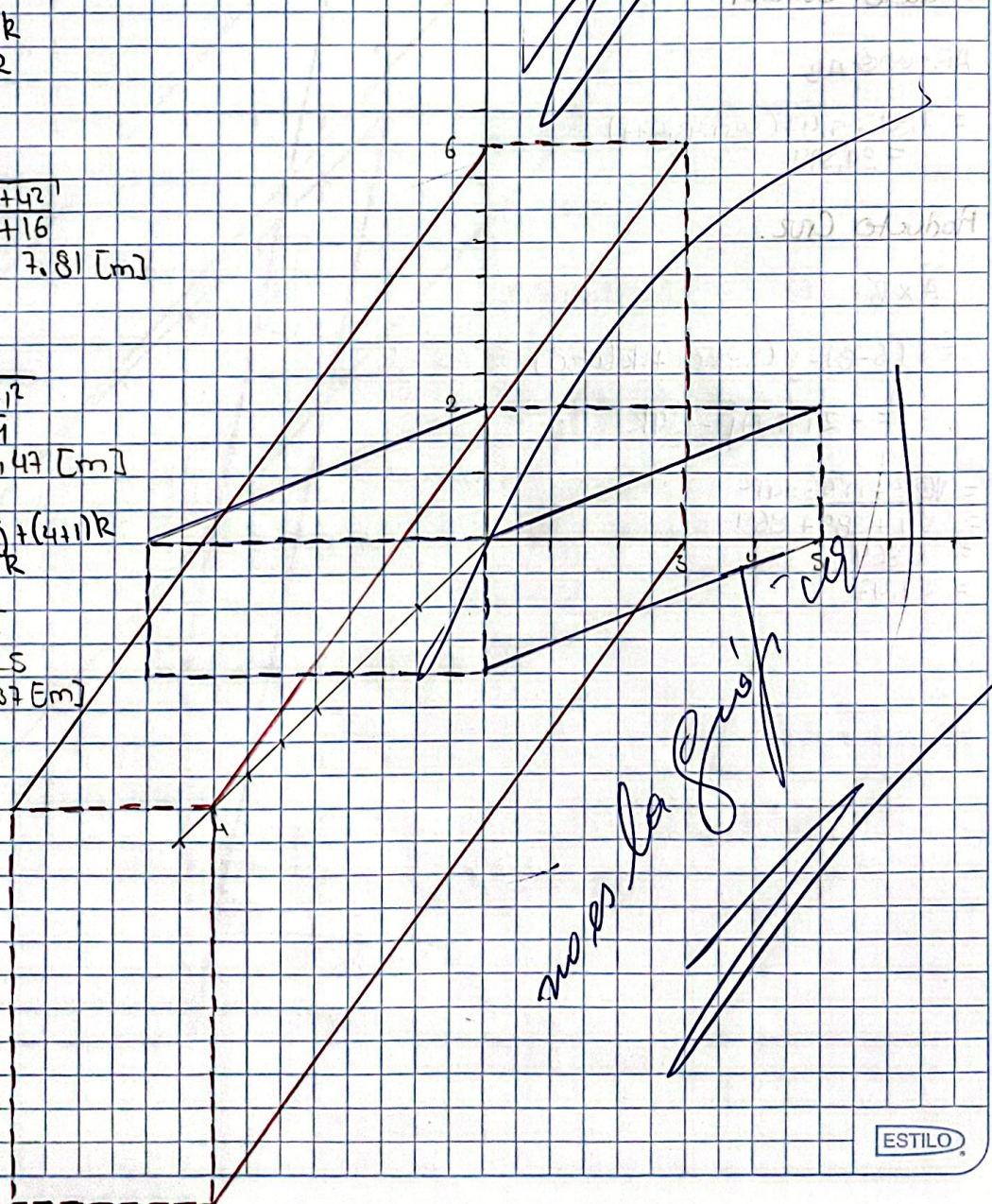
$$\vec{R} = (3+5)\vec{i} + (6+2)\vec{j} + (4+1)\vec{k}$$

$$\vec{R} = 8\vec{i} + 8\vec{j} + 5\vec{k}$$

$$|\vec{R}| = \sqrt{8^2 + 8^2 + 5^2}$$

$$|\vec{R}| = \sqrt{64 + 64 + 25}$$

$$|\vec{R}| = \sqrt{153} = 12.37 \text{ [m]}$$



ESTILO

Producto Escalar

Ángulo

$$|\vec{A}| = 7,81 \text{ [m]}$$

$$|\vec{B}| = 5,47 \text{ [m]}$$

$$|\vec{A} \cdot \vec{B}| = (3)(5) + (6)(2) + (4)(1)$$

$$|\vec{A} \cdot \vec{B}| = 15 + 12 + 4$$

$$|\vec{A} \cdot \vec{B}| = 31$$

$$\theta = \cos^{-1} \left(\frac{31}{(7,81)(5,47)} \right)$$

$$\theta = 43,477$$

Producto Escalar

AB sen θ AB

$$= 7,81 \cdot 5,47 (\sin 43,477)$$

$$= 29,394$$

Producto Cruz.

$$\vec{A} \times \vec{B}$$

$$= i(6 \cdot 8) - j(3 \cdot 20) + k(6 \cdot 30)$$

$$= -2i + 17j - 24k$$

$$= \sqrt{6^2 + 17^2 + 24^2}$$

$$= \sqrt{4 + 289 + 869}$$

$$= \sqrt{869}$$

$$= 29,47$$

• Grupo 6

$$\vec{A} = 3\vec{i} + 4\vec{j} + 6\vec{k}$$

$$\vec{B} = -5\vec{i} + 6\vec{j} + 8\vec{k}$$

• Módulo A

$$|\vec{A}| = \sqrt{3^2 + 4^2 + 6^2}$$

$$|\vec{A}| = \sqrt{9 + 16 + 36}$$

$$|\vec{A}| = \sqrt{61}$$

$$= 7.81 \text{ [m]}$$

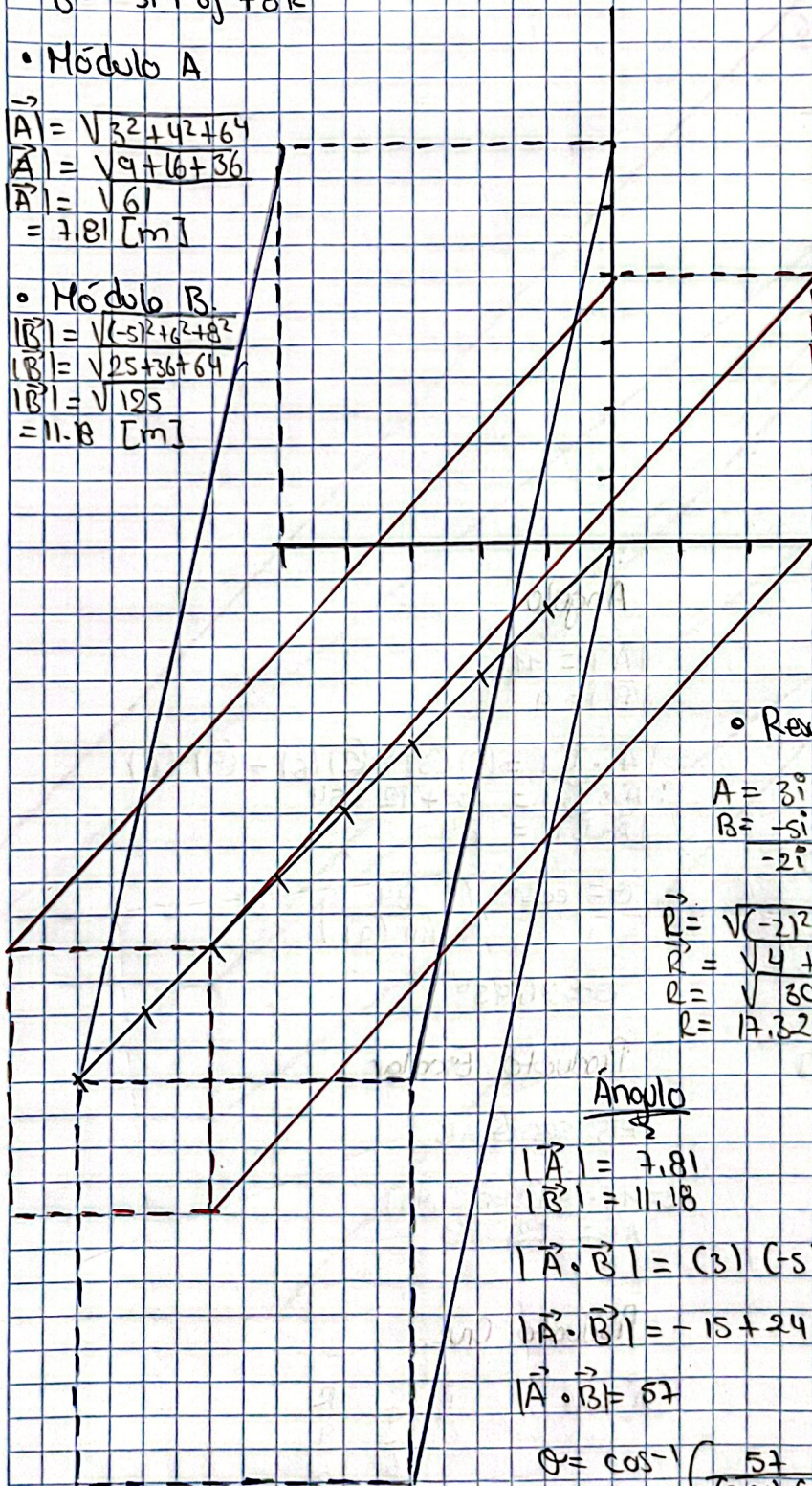
• Módulo B

$$|\vec{B}| = \sqrt{(-5)^2 + 6^2 + 8^2}$$

$$|\vec{B}| = \sqrt{25 + 36 + 64}$$

$$|\vec{B}| = \sqrt{125}$$

$$= 11.18 \text{ [m]}$$



• Resultante.

$$\vec{A} = 3\vec{i} + 4\vec{j} + 6\vec{k}$$

$$\vec{B} = -5\vec{i} + 6\vec{j} + 8\vec{k}$$

$$-\vec{B} = 5\vec{i} - 6\vec{j} - 8\vec{k}$$

$$\vec{R} = \vec{A} + (-\vec{B}) = 8\vec{i} - 2\vec{j} - 2\vec{k}$$

$$|\vec{R}| = \sqrt{8^2 + (-2)^2 + (-2)^2}$$

$$|\vec{R}| = \sqrt{64 + 4 + 4}$$

$$|\vec{R}| = \sqrt{72}$$

$$R = 8.49$$

• Ángulo

$$|\vec{A}| = 7.81$$

$$|\vec{B}| = 11.18$$

$$|\vec{A} \cdot \vec{B}| = (3)(-5) + (4)(6) + (6)(8)$$

$$|\vec{A} \cdot \vec{B}| = -15 + 24 + 48$$

$$|\vec{A} \cdot \vec{B}| = 57$$

$$\theta = \cos^{-1} \left(\frac{57}{(7.81)(11.18)} \right)$$

$$\theta = 49.246$$

◦ Producto Escalar

$$AB \text{ sen } \theta \text{ AB}$$

$$= 7,81 \cdot 11,18 (\text{sen } 49,246)$$

$$= 66,143$$

◦ Producto Cruz

$$\vec{A} \times \vec{B}$$

$$= -4\mathbf{i} - 54\mathbf{j} + 38\mathbf{k}$$

$$= \sqrt{(-4)^2 + (-54)^2 + 38^2}$$

$$= \sqrt{16 + 2916 + 1444}$$

$$= \sqrt{4376}$$

$$= 66,151$$

• Grupo 2

$$\vec{A} = 6\mathbf{i} + 2\mathbf{j} + 9\mathbf{k}$$

$$\vec{B} = 3\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}$$

Módulo $\vec{A}$

$$|\vec{A}| = \sqrt{6^2 + 2^2 + 9^2}$$

$$|\vec{A}| = \sqrt{36 + 4 + 81}$$

$$|\vec{A}| = \sqrt{121} = 11 [\text{m}]$$

Módulo $\vec{B}$

$$|\vec{B}| = \sqrt{3^2 + 6^2 + 6^2}$$

$$|\vec{B}| = \sqrt{9 + 36 + 36}$$

$$|\vec{B}| = \sqrt{81} = 9 [\text{m}]$$

- Resultante

$$\begin{array}{r} 6\mathbf{i} + 2\mathbf{j} + 9\mathbf{k} \\ 3\mathbf{i} + 6\mathbf{j} + 6\mathbf{k} \\ \hline 9\mathbf{i} + 8\mathbf{j} + 15\mathbf{k} \end{array}$$

$$R = \sqrt{9^2 + 8^2 + 15^2}$$

$$R = \sqrt{81 + 64 + 225}$$

$$R = \sqrt{370} = 19,23$$

Ángulo

$$|\vec{A}| = 11$$

$$|\vec{B}| = 9$$

$$|\vec{A} \cdot \vec{B}| = (6)(3) + (2)(6) + (9)(6)$$

$$|\vec{A} \cdot \vec{B}| = 18 + 12 + 54$$

$$|\vec{A} \cdot \vec{B}| = 84$$

$$\theta = \cos^{-1} \left(\frac{84}{(11)(9)} \right)$$

$$\theta = 31,95^\circ$$

Producto Escalar

$$AB \text{ sen } \theta \text{ AB}$$

$$AB = 11 \cdot 9 (\text{sen } 31,95)$$

$$AB = 52,388$$

Producto Cruz

$$\vec{A} \times \vec{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 2 & 9 \\ 3 & 6 & 6 \end{vmatrix}$$

$$\vec{A} \times \vec{B} = -42\mathbf{i} - 9\mathbf{j} + 30\mathbf{k}$$

$$= \sqrt{(-42)^2 + 9^2 + 30^2}$$

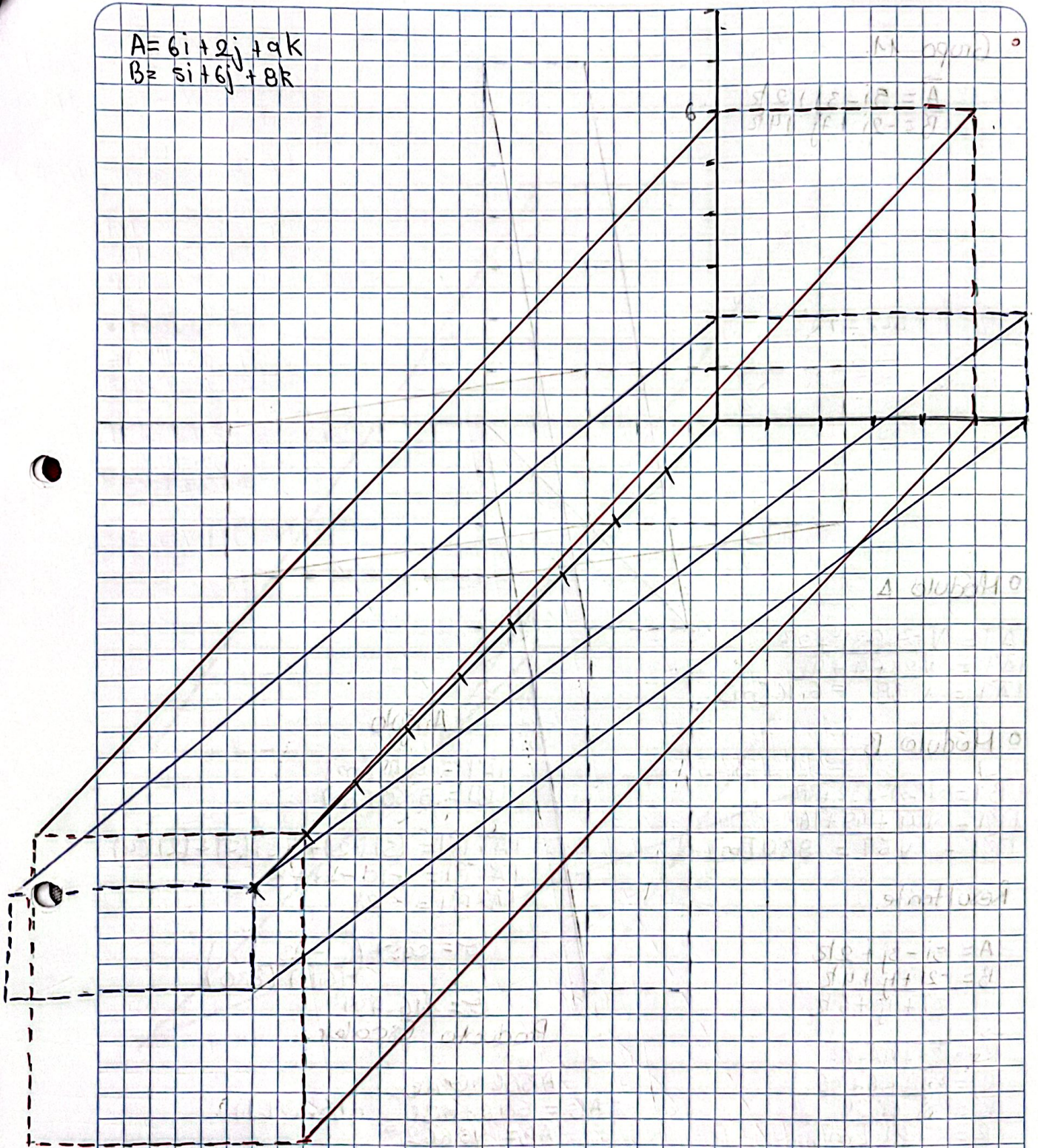
$$= \sqrt{1764 + 81 + 900}$$

$$= \sqrt{2745}$$

$$= 52,39$$

$$A = 6i + 2j + 9k$$

$$B = 5i + 6j + 8k$$



ESTILO

• Grupo 11.

$$\vec{A} = 5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$$

$$\vec{B} = -2\mathbf{i} + 7\mathbf{j} + 4\mathbf{k}$$

o Módulo A

$$|\vec{A}| = \sqrt{5^2 + (-3)^2 + 2^2}$$

$$|\vec{A}| = \sqrt{25 + 9 + 4}$$

$$|\vec{A}| = \sqrt{38} = 6,16 [\text{m}]$$

o Módulo B

$$|\vec{B}| = \sqrt{(-2)^2 + 7^2 + 4^2}$$

$$|\vec{B}| = \sqrt{4 + 49 + 16}$$

$$|\vec{B}| = \sqrt{69} = 8,30 [\text{m}]$$

Resultante.

$$A = 5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$$

$$B = -2\mathbf{i} + 7\mathbf{j} + 4\mathbf{k}$$

$$3\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$$

$$\vec{R} = \sqrt{3^2 + 4^2 + 6^2}$$

$$\vec{R} = \sqrt{9 + 16 + 36}$$

$$\vec{R} = \sqrt{61}$$

$$R = 7,81 [\text{m}]$$

$$= \sqrt{624^2 + (-24)^2 + 29^2}$$

$$= \sqrt{626 + 576 + 841}$$

$$= \sqrt{2093}$$

$$= 45,74$$

Ángulo

$$|\vec{A}| = 6,16 [\text{m}]$$

$$|\vec{B}| = 8,30 [\text{m}]$$

$$\vec{A} \cdot \vec{B} = (5)(-2) + (-3)(7) + (2)(4)$$

$$\vec{A} \cdot \vec{B} = -10 - 21 + 8$$

$$\vec{A} \cdot \vec{B} = -23$$

$$\theta = \cos^{-1} \left(\frac{-23}{(6,16)(8,30)} \right)$$

$$\theta = 116,734^\circ$$

Producto Escalar.

$$AB \cos \theta$$

$$AB = 6,16 \cdot 8,30 (\cos 116,734^\circ)$$

$$AB = 45,662^\circ$$

Producto Cruz.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & -3 & 2 \\ -2 & 7 & 4 \end{vmatrix}$$

$$\vec{A} \times \vec{B} = -26\mathbf{i} - 24\mathbf{j} + 29\mathbf{k}$$

• Grupo 10

$$\vec{A} = 12\hat{i} - 5\hat{j} + 15\hat{k}$$

$$\vec{B} = 11\hat{i} + 18\hat{j} + 4\hat{k}$$

• Módulo A

$$|\vec{A}| = \sqrt{12^2 + (-5)^2 + 15^2}$$

$$|\vec{A}| = \sqrt{144 + 25 + 225}$$

$$|\vec{A}| = \sqrt{394} = 19.85 \text{ [m]}$$

• Módulo B

$$|\vec{B}| = \sqrt{11^2 + 18^2 + 4^2}$$

$$|\vec{B}| = \sqrt{121 + 324 + 16}$$

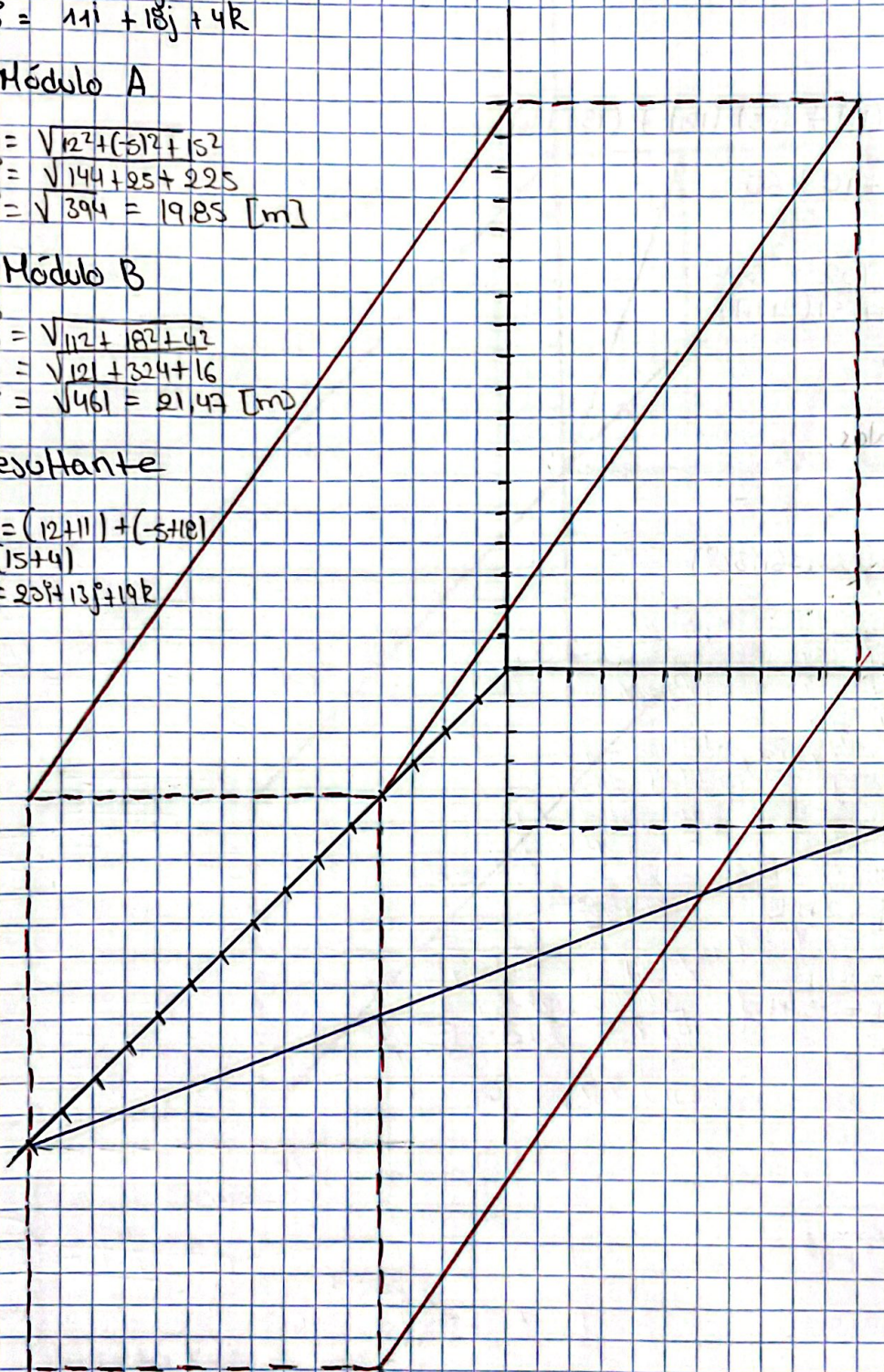
$$|\vec{B}| = \sqrt{461} = 21.47 \text{ [m]}$$

Resultante

$$\vec{R} = (12+11)\hat{i} + (-5+18)\hat{j}$$

$$+ (15+4)\hat{k}$$

$$\vec{R} = 23\hat{i} + 13\hat{j} + 19\hat{k}$$



P_r

Ángulo

$$|\vec{A}| = 19,85$$

$$|\vec{B}| = 21,47$$

$$|\vec{A} \cdot \vec{B}| = (12)(11) + (-5)(18) + (15)(4)$$

$$|\vec{A} \cdot \vec{B}| = 132 - 90 + 60$$

$$|\vec{A} \cdot \vec{B}| = 102$$

$$\theta = \cos^{-1} \left(\frac{102}{(19,85)(21,47)} \right)$$

$$\theta = 76,182^\circ$$

Producto Escalar.

AB sen θ AB

$$= 19,85 \cdot 21,47 (\sin 76,182^\circ)$$

$$= 413,792$$

Producto cruz.

$$\vec{A} \times \vec{B}$$

$$= \begin{vmatrix} i & j & k \\ 12 & -5 & 15 \\ 11 & 18 & 4 \end{vmatrix}$$

$$= -290i + 117j + 271k$$

$$= \sqrt{(-290)^2 + 117^2 + 271^2}$$

$$= \sqrt{84 + 13,689 + 73,441}$$

$$= \sqrt{171,23}$$

$$= 413,792$$

• Grupo 9

$$\vec{A} = 5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$$

$$\vec{B} = 3\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$$

- Módulo A

$$|\vec{A}| = \sqrt{5^2 + 5^2 + 5^2}$$

$$|\vec{A}| = \sqrt{25 + 25 + 25}$$

$$|\vec{A}| = \sqrt{75} = 8,66 [\text{m}]$$

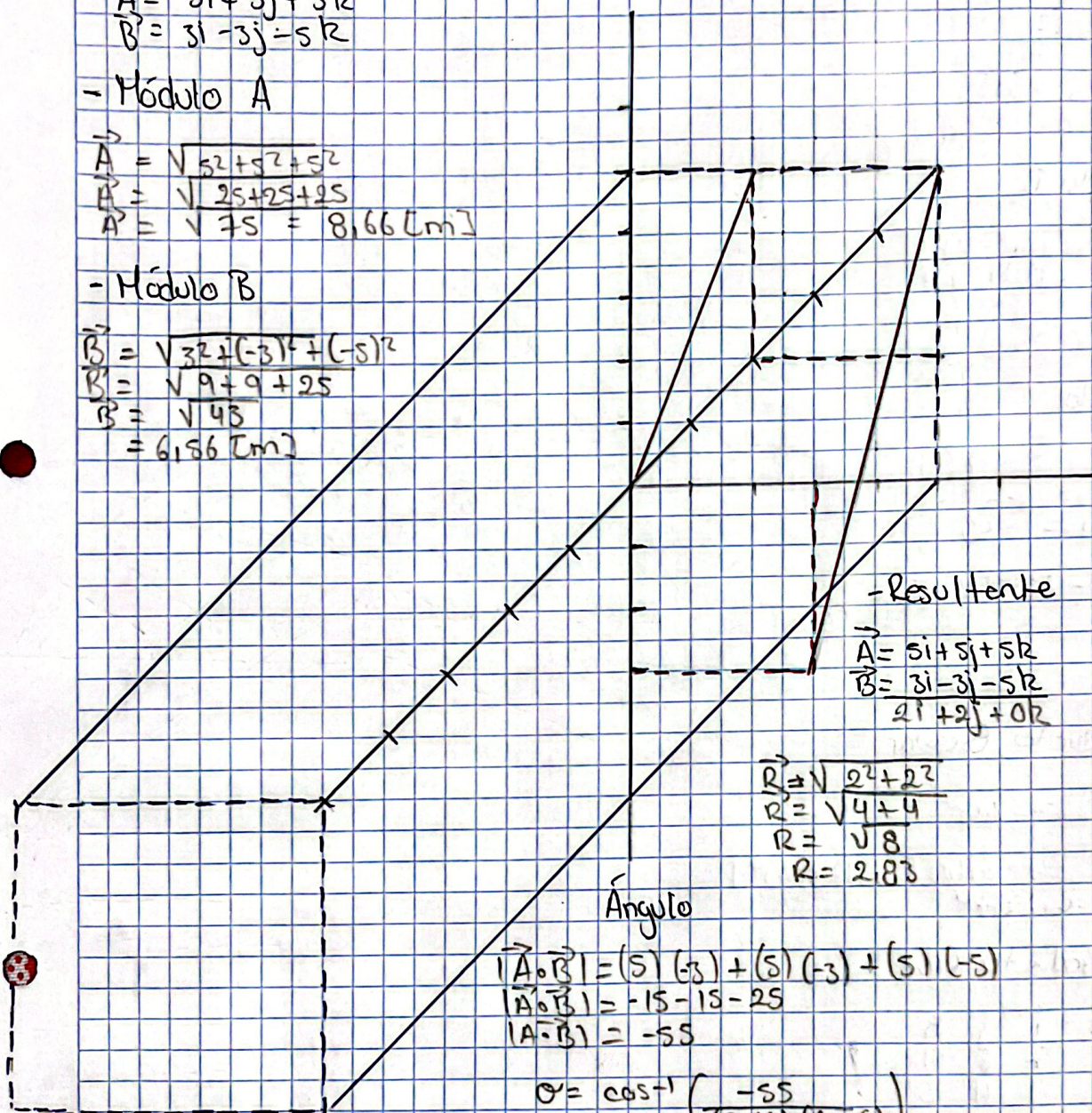
- Módulo B

$$|\vec{B}| = \sqrt{3^2 + (-3)^2 + (-5)^2}$$

$$|\vec{B}| = \sqrt{9 + 9 + 25}$$

$$|\vec{B}| = \sqrt{43}$$

$$|\vec{B}| = 6,56 [\text{m}]$$



- Resultante

$$\vec{A} = 5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$$

$$\vec{B} = 3\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$$

$$2\mathbf{i} + 2\mathbf{j} + 0\mathbf{k}$$

$$|\vec{R}| = \sqrt{2^2 + 2^2}$$

$$|\vec{R}| = \sqrt{4 + 4}$$

$$|\vec{R}| = \sqrt{8}$$

$$|\vec{R}| = 2,83$$

Ángulo

$$|\vec{A} \cdot \vec{B}| = (5)(3) + (5)(-3) + (5)(-5)$$

$$|\vec{A} \cdot \vec{B}| = 15 - 15 - 25$$

$$|\vec{A} \cdot \vec{B}| = -55$$

$$\theta = \cos^{-1} \left(\frac{-55}{(8,66)(6,56)} \right)$$

$$\theta = 165,499$$

Producto Escalar.

$AB \cos \theta$

$$AB = 8,66 \cdot 6,56 (\cos 165,499)$$

$$AB = 14,2240$$

Producto cruz.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 5 & 5 \\ 3 & -3 & -5 \end{vmatrix}$$

$$\vec{A} \times \vec{B} = -10 + 10\mathbf{j}$$

$$= \sqrt{(-10)^2 + 10^2}$$

$$= \sqrt{100 + 100}$$

$$= \sqrt{200}$$

$$= 14,14$$

ESTILO

$$\vec{A} = -7\mathbf{i} + 6\mathbf{j} + 4\mathbf{k} \text{ [m]}$$

$$\vec{B} = 4\mathbf{i} - 4\mathbf{j} + 3\mathbf{k} \text{ [m]}$$

$$|\vec{A}| = \sqrt{7^2 + 6^2 + 4^2}$$

$$|\vec{B}| = \sqrt{4^2 + 3^2 + 16}$$

$$|\vec{A}| = \sqrt{101}$$

$$|\vec{A}| = 10,04 \text{ [m]}$$

Módulo B

$$|\vec{B}| = \sqrt{4^2 + 16 + 9}$$

$$|\vec{B}| = \sqrt{16 + 16 + 9}$$

$$|\vec{B}| = \sqrt{41}$$

$$= 6,40 \text{ [m]}$$

Ángulo

$$\vec{A} \cdot \vec{B} = (-7)(4) + (6)(-4) + (4)(3)$$

$$\vec{A} \cdot \vec{B} = -28 - 24 + 12$$

$$\vec{A} \cdot \vec{B} = -40$$

$$\theta = \cos^{-1} \left(\frac{-40}{(10,04)(6,40)} \right)$$

$$\theta = 128,50^\circ$$

Producto Escalar

AB sen θ AB

$$= 10,04 \cdot 6,40 (\sin 128,50^\circ)$$

$$= 50,287^\circ$$

Producto Cruz.

$$\vec{A} \times \vec{B} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -7 & 6 & 4 \\ 4 & -4 & 3 \end{vmatrix}$$

$$\vec{A} \times \vec{B} = 34\mathbf{i} + 37\mathbf{j} + 4\mathbf{k}$$

$$= \sqrt{34^2 + 37^2 + 4^2}$$

$$= \sqrt{1156 + 1369 + 16}$$

$$= \sqrt{2541}$$

$$= 50,41$$